Create a program for the following conditions with **try, except, else,** and **finally**. Use Replit to see if your code is working or not.

1. **A program that counts the numbers of even and odd numbers of a given range.**

```
try:
  first_int = int(input("Enter the start of the range: "))
  second_int = int(input("Enter the end of the range: "))

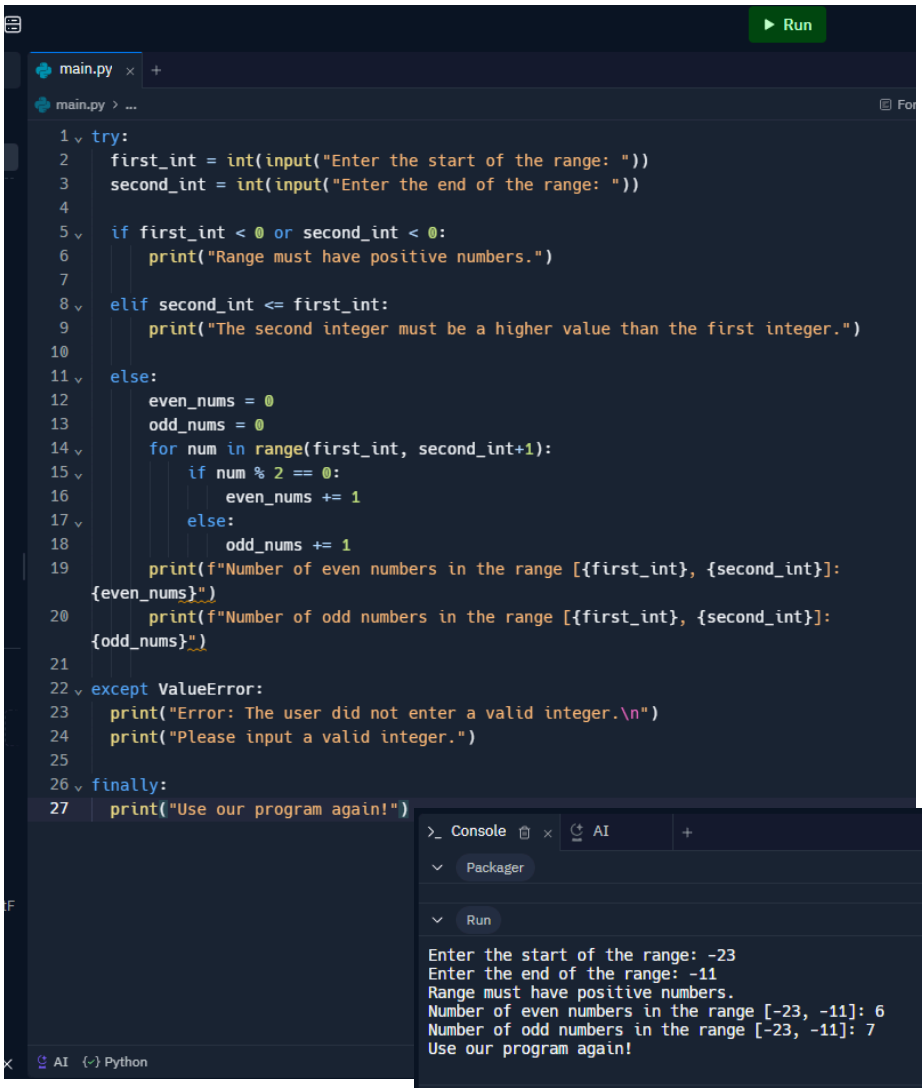
  if first_int < 0 or second_int < 0:
      print("Range must have positive numbers.")

  elif second_int <= first_int:
      print("The second integer must be a higher value than the first integer.")

  else:
      even_nums = 0
      odd_nums = 0
      for num in range(first_int, second_int+1):
          if num % 2 == 0:
              even_nums += 1
          else:
              odd_nums += 1
      print(f"Number of even numbers in the range [{first_int}, {second_int}]: {even_nums}")
      print(f"Number of odd numbers in the range [{first_int}, {second_int}]: {odd_nums}")

except ValueError:
  print("Error: The user did not enter a valid integer.\n")
  print("Please input a valid integer.")

finally:
  print("Use our program again!")
```



```python
try:
    first_int = int(input("Enter the start of the range: "))
    second_int = int(input("Enter the end of the range: "))

    if first_int < 0 or second_int < 0:
        print("Range must have positive numbers.")

    elif second_int <= first_int:
        print("The second integer must be a higher value than the first integer.")

    else:
        even_nums = 0
        odd_nums = 0
        for num in range(first_int, second_int+1):
            if num % 2 == 0:
                even_nums += 1
            else:
                odd_nums += 1
        print(f"Number of even numbers in the range [{first_int}, {second_int}]: {even_nums}")
        print(f"Number of odd numbers in the range [{first_int}, {second_int}]: {odd_nums}")

except ValueError:
    print("Error: The user did not enter a valid integer.\n")
    print("Please input a valid integer.")

finally:
    print("Use our program again!")
```

```
Console  AI

Packager                                   3s on 02:04:08, 03/20 ✓

Run                                        4s on 02:04:11, 03/20 ✓

Enter the start of the range: -23
Enter the end of the range: -11
Range must have positive numbers.
Number of even numbers in the range [-23, -11]: 6
Number of odd numbers in the range [-23, -11]: 7
Use our program again!

Packager                                   3s on 02:06:33, 03/20 ✓

--> poetry lock --no-update
Resolving dependencies...

Run                                        2s on 02:06:36, 03/20 ✓

Enter the start of the range: 22
Enter the end of the range: 11
The second integer must be a higher value than the first integer.
Use our program again!

Packager                                   2s on 02:06:51, 03/20 ✓

--> poetry lock --no-update
Resolving dependencies...

Run                                        1s on 02:06:54, 03/20 ✓

Enter the start of the range: aa
Error: The user did not enter a valid integer.

Please input a valid integer.
Use our program again!

Packager                                   3s on 02:06:58, 03/20 ✓

--> poetry lock --no-update
Resolving dependencies...

Run                                        3s on 02:07:01, 03/20 ✓

Enter the start of the range: -22
Enter the end of the range: -31
Range must have positive numbers.
Use our program again!
```

1. **A program that gives grade based on the grading criteria:**
    - **A: 90% +**
    - **B: 80% - 89%**
    - **C: 70% - 79%**
    - **D: 60% - 69%**
    - **E: 50% - 59%**

```
def ggrado(score):
  if score >= 90:
     return "A"
  elif score >= 80:
     return "B"
  elif score >= 70:
     return "C"
  elif score >= 60:
     return "D"
  elif score >= 50:
     return "E"
  else:
     return "F"

NUM = 2
while NUM == 2:
    def main():
      try:
         score = float(input("Enter your score: "))
         if score < 0 or score > 100:
            raise ValueError("Score must be between 0 and 100.")
         grade = ggrado(score)
         print(f"\nYour grade is: {grade}\n")
      except ValueError as e:
         print(f"Invalid user input. Please enter a score between 0 and 100: {e}")
      except Exception as e:
         print(f"An error has occurred: {e}")
      else:
         print("No errors occurred.")
      finally:
         print("Use this program again!\n")

    if __name__ == "__main__":
      main()
```

main.py

```python
def ggrado(score):
    if score >= 90:
        return "A"
    elif score >= 80:
        return "B"
    elif score >= 70:
        return "C"
    elif score >= 60:
        return "D"
    elif score >= 50:
        return "E"
    else:
        return "F"

NUM = 2
while NUM == 2:
    def main():
        try:
            score = float(input("Enter your score: "))
            if score < 0 or score > 100:
                raise ValueError("Score must be between 0 and 100.")
            grade = ggrado(score)
            print(f"\nYour grade is: {grade}\n")
        except ValueError as e:
            print(f"Invalid user input. Please enter a score between 0 and 100: {e}")
        except Exception as e:
            print(f"An error has occurred: {e}")
        else:
            print("No errors occurred.")
        finally:
            print("Use this program again!\n")

    if __name__ == "__main__":
        main()
```

Console

```
Enter your score: 22

Your grade is: F

No errors occurred.
Use this program again!

Enter your score: 66

Your grade is: D

No errors occurred.
Use this program again!

Enter your score: 99

Your grade is: A

No errors occurred.
Use this program again!

Enter your score: 88

Your grade is: B

No errors occurred.
Use this program again!

Enter your score: 77

Your grade is: C

No errors occurred.
Use this program again!

Enter your score: 66

Your grade is: D

No errors occurred.
Use this program again!

Enter your score: 55

Your grade is: E

No errors occurred.
Use this program again!
```